

Transforming Critical Care: Machine Learning Models for Real-Time AKI Prediction

The Problem: A Clinical Dilemma

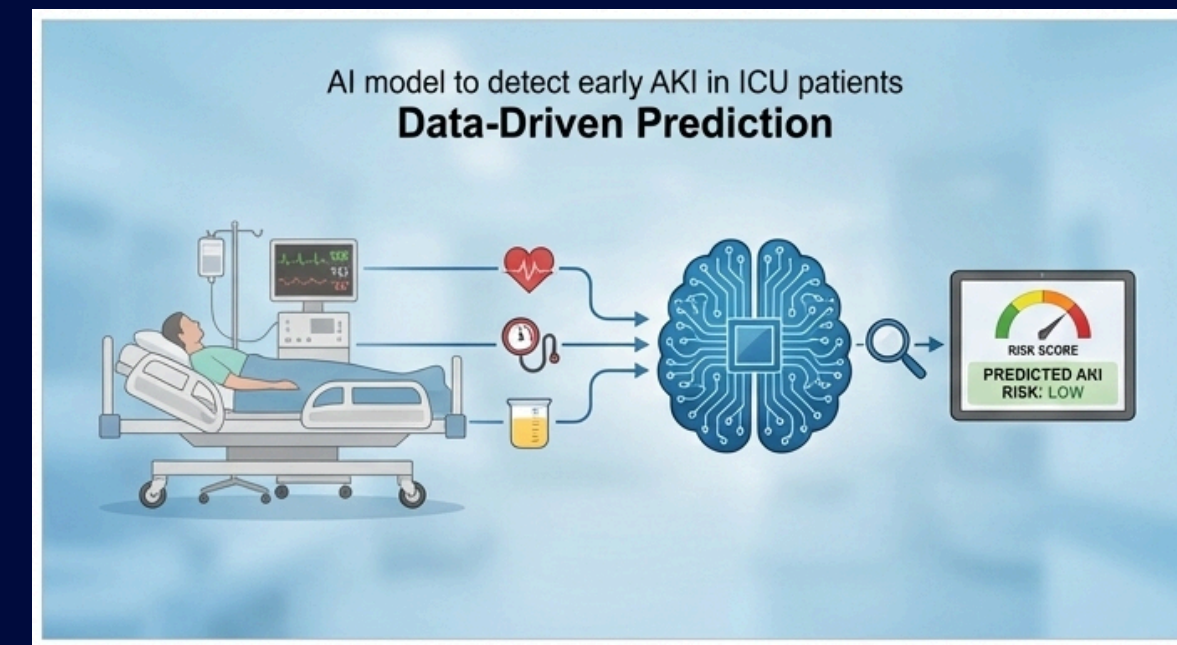
- AKI affects up to 50% of ICU patients, increasing mortality and costs.
- Lagging Indicators: Current gold standards (Creatinine/Urine Output) are reactive; by the time they change, renal damage is already sustained.

Objective:

Develop a high-fidelity machine learning model to forecast AKI in advance, shifting clinical management from reactive treatment to proactive prevention.

Data Integrity & Cleaning (The Cohort):

- Source: MIMIC-IV Database (~75,000 stays cleaned to a high-fidelity adult cohort).
- Cleaning Protocol: Removed "impossible" outliers (e.g., Urine Output > 50L/day, weights < 30kg, Negative Heart Rates).
- Final Feature Set: 46 variables across 7 domains (Labs, Vitals, Fluids, Comorbidities).
- The data was cleaned by handling null values and missing values appropriately.
- Constant sanity checks were made to ensure integrity of model.



The Model

Feature Engineering

Demographics & Admission Type:

Age, sex, emergency vs. elective admission

Laboratory Values:

Creatinine, urea, Na⁺, Cl⁻, CPK, and urine output

Fluid Balance:

Total fluid input and output

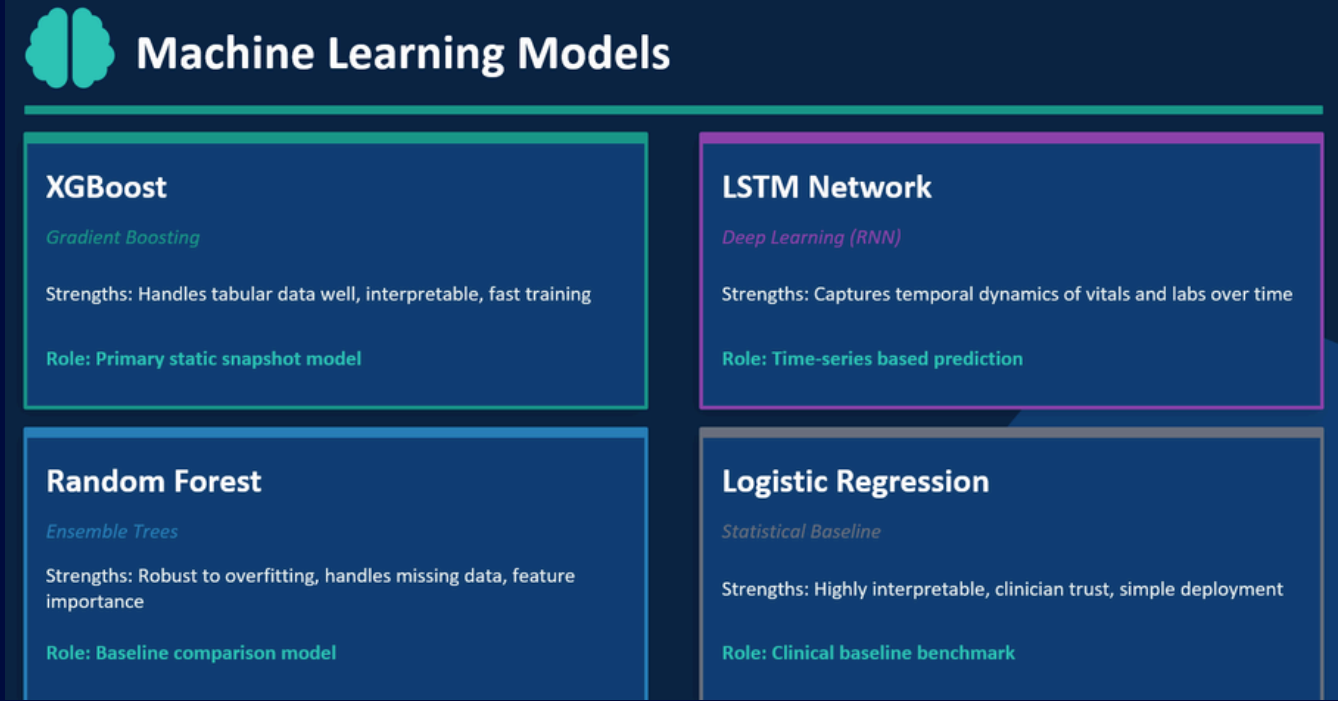
Hemodynamics & Neurologic Status:

Blood pressure, Glasgow Coma Scale (GCS)
SOFA, delta-SOFA Saline Administration

Comorbidities:

Diabetes, hypertension, chronic kidney disease

Model Training



With Clinical Decision Support:

Class imbalance addressed using weighting techniques

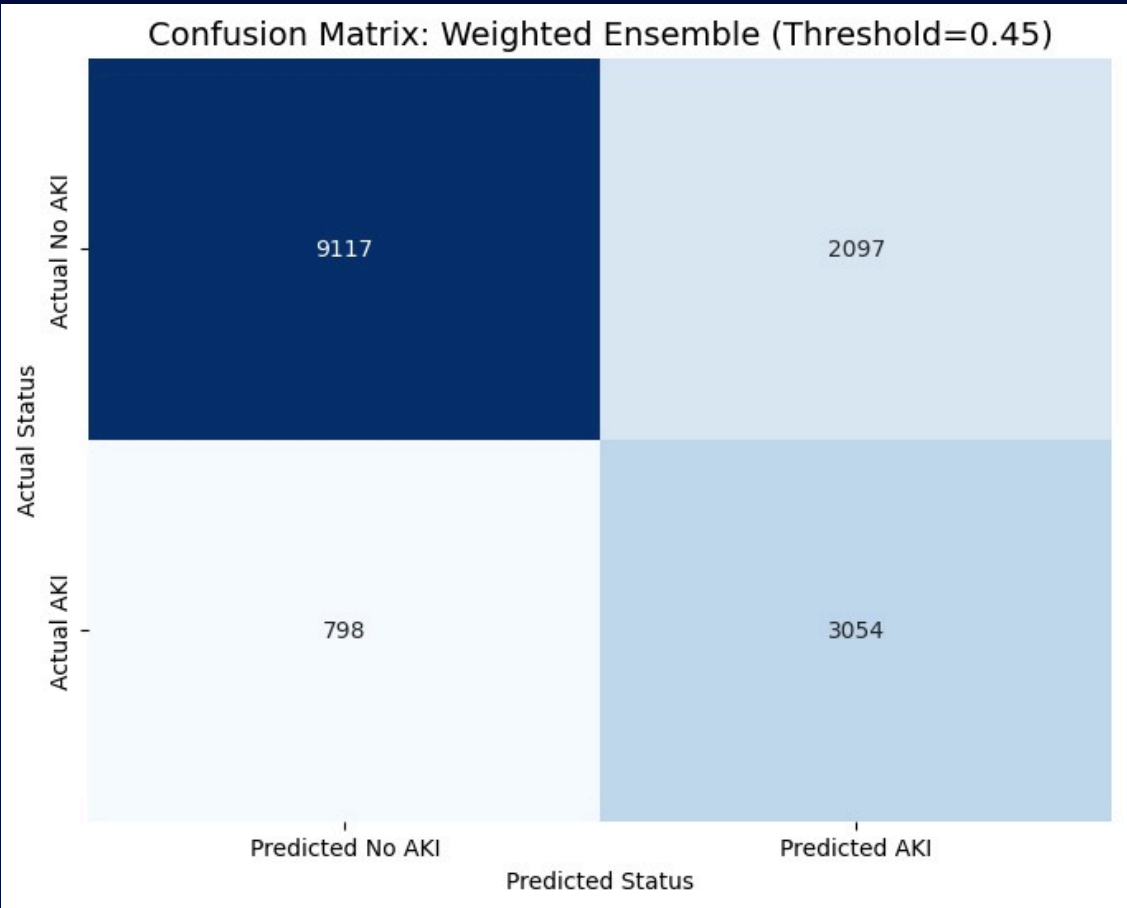
Predictions combined via ensemble learning

Threshold optimized for clinically meaningful outcomes

Observations

Model Impact Report:

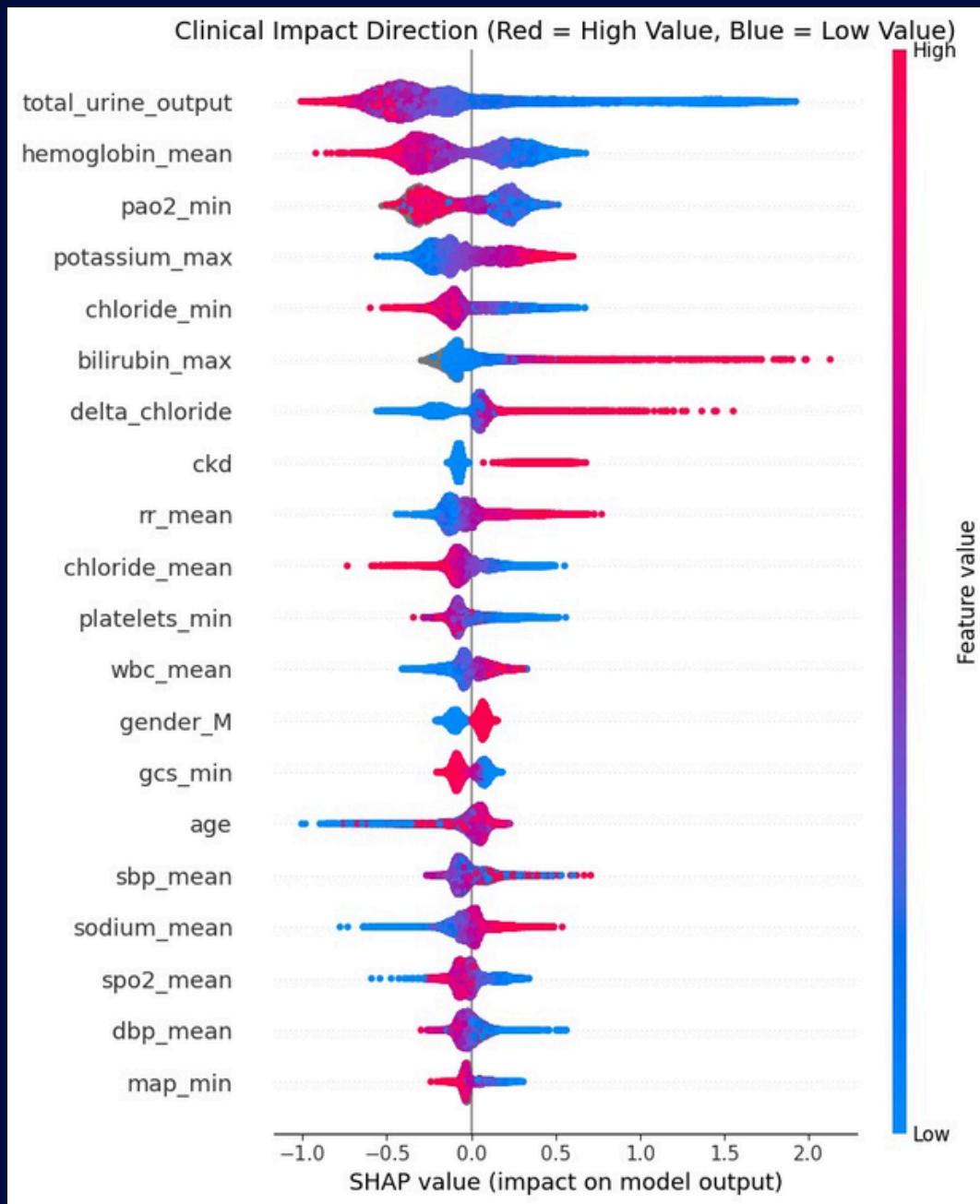
	Metric	Value
	AUROC	0.8884
AUPRC (Precision-Recall)		0.7707
Recall (Sensitivity)		0.8123
Precision		0.5738
F1-Score		0.6725



The Results

- Strongest negative correlation observed urine output in ml/kg-hr whereas the negative correlation for total urine output is not as strong as the former due the patient's weight coming to consideration
- **delta_creatinine** has a strong positive correlation as expected from the patients with AKI.
- **map_mean & delta_map**: These show a negative correlation. As the Mean Arterial Pressure (or the change in pressure) drops, the AKI label appears more frequently.
- **hr_mean (Heart Rate)**: Shows a positive correlation. Higher average heart rates are associated with AKI, likely representing a physiological stress response.

Raw Chloride levels appear to show a negative correlation. Patients at risk for AKI are often aggressively resuscitated with various fluids. If they receive a large amount of non-saline fluids or experience significant third-spacing , their serum chloride can become diluted. What we have looked at the delta chloride levels i.e. the increase in the chloride levels over a 24 hour period which shows that with increase in chloride levels , there is a direct correlation with the onset of AKI.



Key Clinical Observations

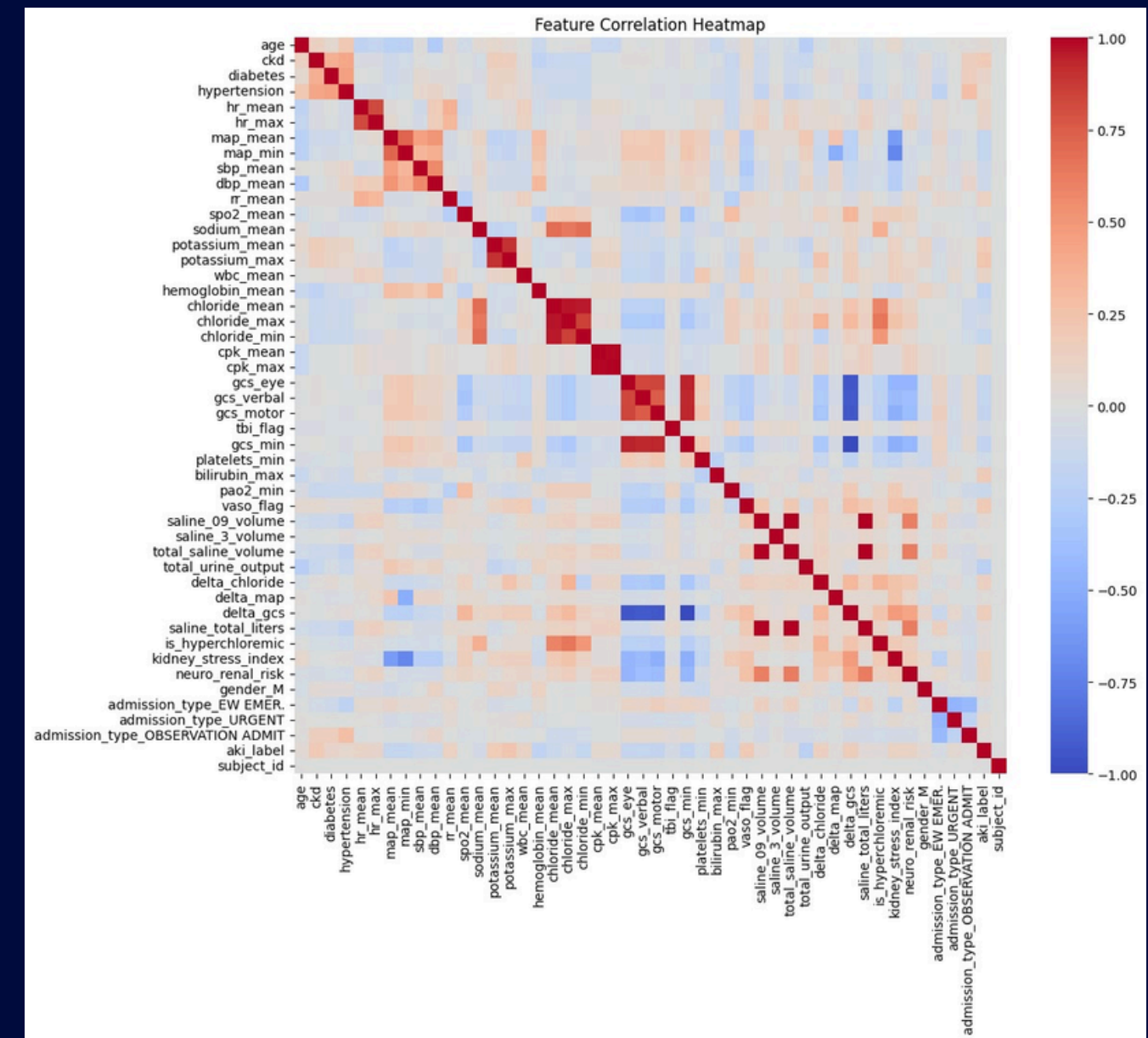
- Total Saline Volume: Higher fluid intake is associated with increased AKI risk
- BUN (Blood Urea Nitrogen): Rising average BUN strongly indicates higher likelihood of AKI
- Age & Comorbidities: Older patients and those with existing conditions show higher risk

Multi-Organ Indicators of AKI

- Increasing SOFA score indicates worsening organ function
- Decreasing GCS reflects neurological decline
- Lower MAP signals cardiovascular instability
- Elevated Bilirubin suggests liver dysfunction
- Presence of Vaso_flag indicates hemodynamic failure/shock

Physiological Interpretation

- CPK & Bilirubin spikes reflect tissue and liver damage, increasing kidney stress
- Vaso_flag (shock state) indicates reduced renal perfusion as blood is redirected to vital organs



Challenges

Conflict: We couldn't build a model on our initial hypothesis "The 3% Hypertonic saline given to TBI patients induce AKI" due to lack of relevant data in the MIMIC-IV dataset.

Resolution: We expanded our horizon and used the entirety ICU data to study the relevance of parameters measured during ICU state.

Conflict: We found that the raw ICU data was noisy. We found impossible records, such as patients with 53 liters of urine output in 24 hours and negative heart rates.

Resolution: We established physiological boundaries (e.g., weights 30–300kg, urine < 10L/day). This ensured our model learned from human biology, not sensor errors.

Conflict: A 75/25 class imbalance meant a model could get "75% Accuracy" by simply predicting "No AKI" for everyone—missing every single sick patient.

Resolution: We discarded "Accuracy" as a target. We used cost-sensitive learning (scale_pos_weight) and tuned our threshold to 0.45 to maximize Recall. We would rather have a false alarm than a missed diagnosis.

Clinical Implementation

- Embed model within the Electronic Health Record (EHR) system
- Real-time monitoring of labs, vitals, urine output
- Ensure model transparency and interpretability
- Maintain patient data privacy
- **Use model as decision support, not replacement for clinical judgment**

Takeaways

- Working with raw data is a very humbling process
- How important is it to have a proper and well-defined pipeline
- How to choose the best model for a certain type of data
- While data science finds the patterns, the clinicians prove the purpose
- Interdisciplinary collaboration – everybody had something to teach and something to learn

Future Scope

- Classify patients into Low / Moderate / High AKI risk
- Trigger automated alerts for high-risk patients
- With the necessary data, establish a clear relationship between chloride levels and onset of AKI
- Early identification of AKI as Transient/Persistent/Acute Kidney Disease/Chronic Kidney Disease